

# KnowSelf

## Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate cognitive resources: act directly, self-reflect, or consult external knowledge — using only ~15% knowledge queries while matching or beating fully-augmented GPT-4o methods.

**Fast  
Thinking**

**Slow  
Thinking**

**Knowledgeable  
Thinking**

[arxiv.org/abs/2504.03553](https://arxiv.org/abs/2504.03553)

# The 'Flood Irrigation' Problem

Current agent training indiscriminately injects knowledge — regardless of situational need

## Trajectory Imitation

e.g., ReAct

Blindly imitates gold action sequences.  
No mechanism to gauge confidence or situational fit.

## Trial-and-Error Refinement

e.g., Reflexion

Applies self-reflection indiscriminately at every step.  
Costly even when the action was already correct.

## Knowledge-Augmented Planning

e.g., KnowAgent, WKM

Injects external domain knowledge at 100% of steps.  
Overkill leads to pattern collapse on novel inputs.

Result: Fragile planning, unnecessary cost, pattern collapse on unexpected inputs.

Humans, by contrast, possess metacognitive self-awareness — they know when to act directly, when to pause and reflect, or when to seek external help.  
KnowSelf brings this capability to LLM agents.

# Three Situational Thinking Modes

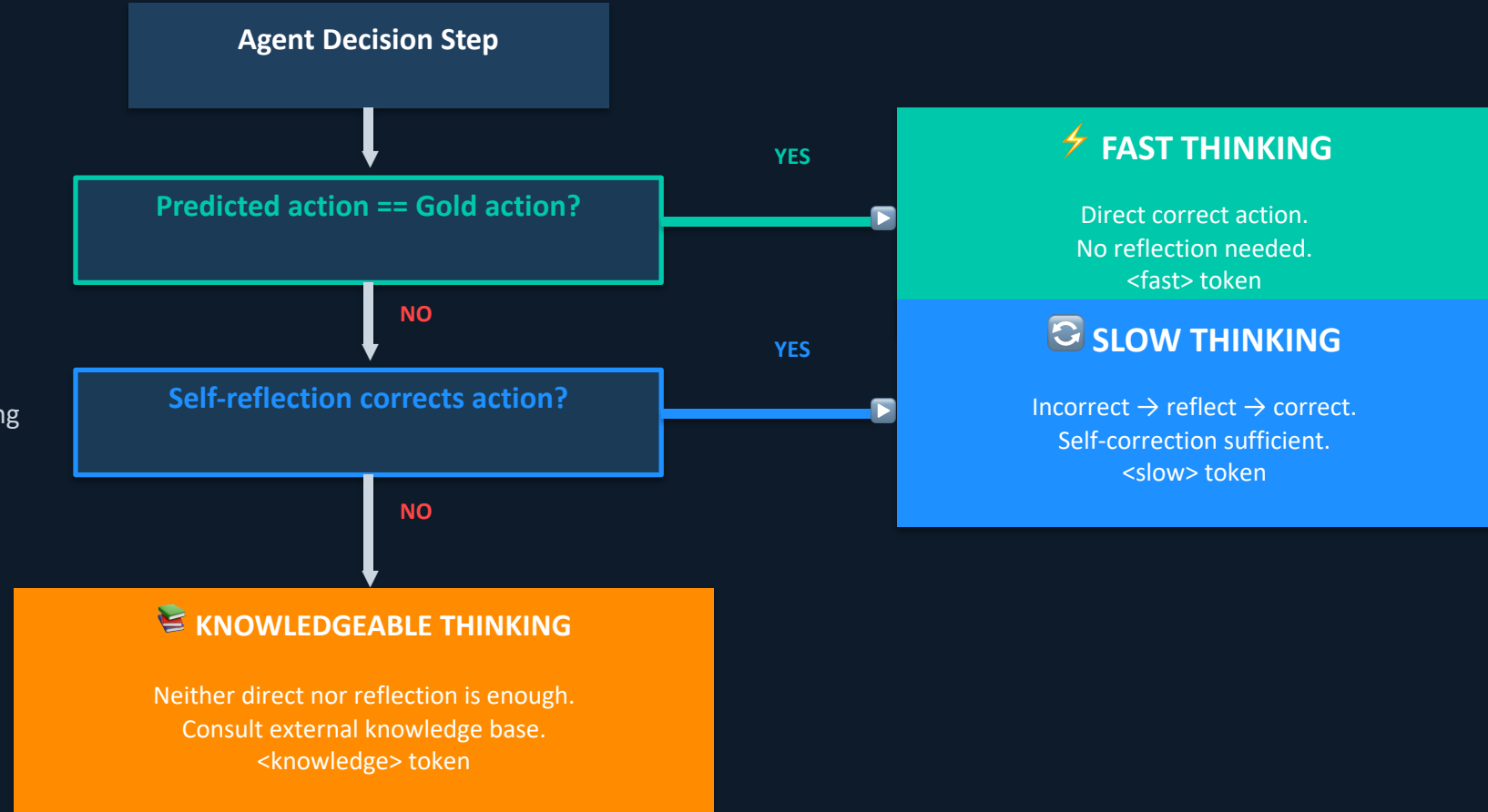
KnowSelf classifies every decision step into one of three cognitive situations

## Decision Criteria

At each planning step, the agent evaluates its own capability:

1. Can I produce the correct action directly?
2. Can I correct myself through reflection?
3. Do I need external knowledge?

Special tokens (<fast>, <slow>, <knowledge>) are inserted during training to mark each situation.



# KnowSelf Pipeline: Three-Stage Framework

Data Construction → Two-Stage Training → Self-Aware Inference

## Step 1

### Self-Awareness Data Construction

- Agent self-explores environments
- Heuristic criterion classifies each step:
  - Match gold action → fast thinking
  - Reflection fixes it → slow thinking
  - Neither works → knowledgeable thinking
- Special tokens inserted into trajectories



## Step 2

### Two-Stage Training

- Stage A — Supervised Fine-Tuning (SFT):  
  
Teaches initial self-awareness patterns  
  
from labeled trajectory data
- Stage B — Reward-based Policy  
  
Optimization (RPO):  
  
Refines special-token generation  
  
via reinforcement signal



## Step 3

### Self-Aware Inference

- Agent generates special tokens at each step
- <fast> → execute action directly
- <slow> → pause, reflect, then act
- <knowledge> → retrieve from external KB,  
then act on augmented context
- Only ~15–26% of steps use knowledge

Reference: KnowSelf Framework Figure (figure\_3) — data construction, SFT+RPO training loop, and token-gated inference.

# KnowSelf in Action: Qualitative Comparisons

Where state-of-the-art LLMs fail, KnowSelf's situational tokens guide correct behavior

## Task A: "Put a hot mug in cabinet"

### ✗ OpenAI-O1 (fails)

Opens microwave → incorrectly tries to take an egg from the microwave. Does not recognize that the open microwave should be used to heat the mug.

### ✓ KnowSelf / Llama-8B [<slow> Reflection token]

Uses <slow> token → reflects that microwave is already open → correctly heats the mug using the open microwave → proceeds to place hot mug in cabinet successfully.

Insight: Slow thinking avoids confabulation; reflection recovers correct plan.

## Task B: "Put a saltshaker in drawer"

### ✗ DeepSeek-R1 (fails)

Reflects that the drawer is already open → incorrectly tries to close the drawer first. This action fails, blocking task completion.

### ✓ KnowSelf / Llama-8B [<knowledge> token]

Uses <knowledge> token → retrieves domain rule: "place the object without removing unrelated items" → places saltshaker directly into open drawer, completing the task.

Insight: Knowledge token provides the missing rule; avoids spurious actions.

# ALFWorld Results: KnowSelf Achieves 84.33% with Only 15% Knowledge Usage

Success rate (%) across six task types. Know% = proportion of steps using external knowledge.

| Backbone | Method    | Know%  | Put    | Clean | Heat  | Cool  | Examine | Put Two | All   |
|----------|-----------|--------|--------|-------|-------|-------|---------|---------|-------|
| GPT-4o   | ReAct     | 0%     | 83.33  | 74.19 | 69.57 | 85.71 | 77.78   | 64.71   | 76.12 |
| GPT-4o   | Reflexion | 0%     | 100.00 | 87.10 | 73.91 | 90.48 | 83.33   | 70.59   | 85.07 |
| GPT-4o   | ExpeL     | 100%   | 95.83  | 83.87 | 69.57 | 80.95 | 88.89   | 52.94   | 79.85 |
| Llama-8B | ReAct     | 0%     | 33.33  | 3.23  | 0.00  | 57.14 | 66.67   | 23.53   | 27.61 |
| Llama-8B | Reflexion | 0%     | 62.96  | 22.73 | 5.88  | 64.29 | 86.36   | 50.00   | 51.49 |
| Llama-8B | ExpeL     | 100%   | 83.33  | 32.26 | 30.43 | 23.81 | 55.56   | 17.65   | 41.04 |
| Llama-8B | ETO       | 0%     | 91.67  | 70.59 | 82.61 | 61.90 | 88.89   | 64.71   | 78.36 |
| Llama-8B | KnowAgent | 100%   | 87.50  | 93.55 | 65.22 | 66.67 | 61.11   | 64.71   | 75.37 |
| Llama-8B | WKM       | 100%   | 95.83  | 87.10 | 86.96 | 61.90 | 66.67   | 52.94   | 77.61 |
| Llama-8B | KnowSelf  | 15.01% | 91.67  | 87.10 | 91.30 | 85.71 | 77.78   | 64.71   | 84.33 |
| Gemma-2B | ReAct     | 0%     | 0.00   | 9.68  | 0.00  | 4.76  | 44.44   | 0.00    | 8.96  |
| Gemma-2B | ETO       | 0%     | 91.67  | 83.87 | 78.26 | 52.38 | 77.78   | 29.41   | 71.64 |
| Gemma-2B | KnowAgent | 100%   | 91.67  | 90.32 | 69.57 | 71.43 | 66.67   | 41.18   | 73.88 |
| Gemma-2B | WKM       | 100%   | 91.67  | 87.10 | 78.26 | 71.43 | 61.11   | 52.94   | 76.12 |
| Gemma-2B | KnowSelf  | 26.41% | 87.50  | 93.55 | 73.91 | 76.19 | 83.33   | 52.94   | 79.85 |

Highlighted rows = KnowSelf entries | Red Know% = 100% (full knowledge injection) | Yellow = best All-task score for that backbone

# Minimal Knowledge, Maximum Impact: Efficiency Advantage

KnowSelf uses 15–26% knowledge queries vs. 100% for competing methods, yet achieves the best accuracy

## Knowledge Query Rate (Know%)



## Overall Accuracy (All Tasks %)



KnowSelf (Llama-8B): 84.33% with 15% knowledge | KnowSelf (Gemma-2B): 79.85% with 26% knowledge | Both beat all 100%-knowledge baselines on same backbone

# Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

KnowSelf demonstrates that knowing when to think is as important as how to think

|   |                                                                                                                                                                                                                                                                        |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <p><b>Indiscriminate Knowledge Injection Is Wasteful and Suboptimal</b></p> <p>All-or-nothing knowledge use (100% Know%) by ExpeL, KnowAgent, and WKM can hurt accuracy. Selective use outperforms blanket augmentation.</p>                                           |
| 2 | <p><b>Agents Can Learn Situational Self-Awareness via Special Tokens</b></p> <p>Lightweight heuristic labeling on self-explored trajectories — no expensive human annotation. &lt;fast&gt;, &lt;slow&gt;, and &lt;knowledge&gt; tokens encode metacognitive state.</p> |
| 3 | <p><b>KnowSelf Beats GPT-4o Methods with Llama-8B</b></p> <p>84.33% on ALFWorld surpasses GPT-4o ReAct (76.12%) and matches/beats GPT-4o ExpeL (79.85%) — using a 8B open-source model.</p>                                                                            |
| 4 | <p><b>Only 15–26% Knowledge Usage Needed</b></p> <p>KnowSelf (Llama-8B) uses 15.01% knowledge; Gemma-2B uses 26.41% — versus 100% for competing knowledge-augmented methods. ~6–7× more efficient.</p>                                                                 |
| 5 | <p><b>Scales Down: Gemma-2B Reaches 79.85% Matching GPT-4o ExpeL</b></p> <p>The self-awareness paradigm generalizes across model scales. Even a 2B-parameter model matches a full-knowledge GPT-4o method.</p>                                                         |

Future directions: extending to multi-hop reasoning, richer knowledge bases, and online adaptation of self-awareness thresholds.